DOCUMENTATION

DATA COLLECTION

The data was collected from the website Project Gutenberg (https://www.gutenberg.org), which provides free access to a large collection of e-books and their metadata. The main goal was to gather structured information about books for further analysis.
The data collection process was implemented in Python using the requests library to send HTTP requests and BeautifulSoup to parse the HTML content. This method was chosen because the website structure is static, making it efficient to extract data without using browser automation tools.
Pagination was handled using the start_index parameter, allowing the scraper to move through multiple pages and collect a large number of records. A custom User-Agent header was added to simulate a real browser request, and small random delays were introduced between requests to reduce server load.

Each book page was parsed to extract important attributes such as title, author, language, subject, category, release date, number of downloads, and source URL. Additionally, a timestamp was recorded for each entry. The collected data was stored in a structured format and converted into a Pandas DataFrame.

To ensure reliability, the scraper includes error handling and logging, which help track progress and manage potential request or parsing issues. Finally, the dataset was saved as a CSV file (gutenberg_raw_5000.csv) for further processing and analysis

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Overall, this approach allowed us to efficiently collect a large volume of structured data in a reliable and scalable way.

DATA CLEANING & TRANSFORMATION

The dataset obtained after the scraping process contained around 5,000 records with variables such as title, author, language, downloads, and release date. Although the dataset was relatively complete, several preprocessing steps were required to improve its quality and consistency.

One of the first issues identified was missing values. The author field was filled with “Unknown” where data was missing. However, records with missing title or release date were removed, since these fields are essential for further analysis.

Another important step was removing duplicate records. Entries with the same title and author were considered duplicates. Keeping such records could distort the analysis, so duplicates were removed to ensure that each book appears only once in the dataset.

After that, the dataset was standardized. The downloads column was converted into numeric format, and the release date was transformed into a proper datetime format. Text columns such as language and category were cleaned by removing extra spaces and ensuring consistency.

Outliers in the dataset were identified using the IQR method, focusing on the downloads variable. Instead of removing these values, they were marked, since highly downloaded books may represent important trends and should not be lost.

Further preprocessing included feature engineering. Several new variables were created to enhance analysis. These include:

- Year (extracted from release date)
- Days Online (how long the book has been available)
- Title Length (number of words in the title)
- Popularity Tier (Bronze, Silver, Gold, Platinum based on downloads)
- Clean Author Name (simplified author format)

These features help transform raw data into more meaningful and analyzable information.

After completing all preprocessing steps, the dataset showed a clear improvement in terms of consistency and usability. Duplicate entries were removed, missing values were properly handled, and all variables were standardized.

In addition, newly created features increased the analytical potential of the dataset, allowing for deeper insights.

Overall, the preprocessing stage was essential in transforming raw scraped data into a clean and well-organized dataset suitable for further analysis.

EXPLORATORY DATA ANALYSIS (EDA)

After preprocessing, the dataset was analyzed using descriptive statistics. Key numerical variables such as downloads, days online, and title length were examined.

Basic statistical measures including mean, median, mode, and quartiles were calculated to understand the central tendency and spread of the data. This step provided a general overview of how the values are distributed across the dataset.

In addition, skewness was measured to evaluate the distribution shape. The results showed that some variables, especially downloads, are not normally distributed and are influenced by highly popular books.

```
# 2. DESCRIPTIVE STATISTICS
# Mean, median, mode, and quartiles
numeric_summary = df[['Downloads', 'Days_Online', 'Title_Length']].describe()
numeric_summary.loc['mode'] = df[['Downloads', 'Days_Online', 'Title_Length']].mode().iloc[0]
print("--- 1. Stats ---\n", numeric_summary)

# 3. SKEWNESS
# Check data distribution
skewness = df[['Downloads', 'Days_Online', 'Title_Length']].skew()
print("\n--- 2. Skewness ---\n", skewness)
```

```
--- 1. Stats ---
            Downloads   Days_Online  Title_Length
count    4853.000000   4853.000000   4853.000000
mean     6000.973831   5516.113744      9.757264
std      6452.678152   2456.191414      3.032255
min      2756.000000     12.000000      1.000000
25%      3357.000000   4066.000000      7.000000
50%      4197.000000   6320.000000     10.000000
75%      6406.000000   6867.000000     12.000000
max    146193.000000  19868.000000     18.000000
mode     2988.000000   6202.000000      9.000000

--- 2. Skewness ---
 Downloads       9.125879
Days_Online    -0.296729
Title_Length    0.097962
dtype: float64
```

To understand relationships between variables, a correlation matrix was created. This helped identify how strongly different features are related to each other.

The analysis focused on variables such as downloads, title length, and days online. Correlation values indicated whether relationships are positive, negative, or weak, helping to identify potential patterns in the data.

Additionally, a group-by analysis was performed based on the popularity tier. The average number of downloads was calculated for each tier, showing clear differences between Bronze, Silver, Gold, and Platinum groups.

To highlight the most successful entries, the top 5 books were selected based on the number of downloads.
This step helped identify the most popular books in the dataset and provided insight into which titles and authors attract the most attention from readers.

```
--- 5. Top 5 Books ---
                                              Title Author_Clean  Downloads
0  Frankenstein; or, the modern prometheus by Mar...      Shelley     146193
1  The City of God, Volume I by Saint of Hippo Au...    Augustine     139919
2        Moby Dick; Or, The Whale by Herman Melville     Melville     118655
3              Pride and Prejudice by Jane Austen       Austen     103210
4           Romeo and Juliet by William Shakespeare  Shakespeare      77436

--- 6. Hypothesis Testing ---
H1 (English vs Others): Confirmed (p=0.0001)
H2 (Title Length): Confirmed (Corr=-0.0468)
H3 (Longevity): Confirmed (Corr=0.1474)
```

Several hypotheses were tested to better understand the factors influencing book popularity.

- H1: English books are more popular than books in other languages
- H2: Title length affects the number of downloads
- H3: Older books (more days online) have higher popularity

Statistical tests such as t-test and Pearson correlation were used. The results showed whether each hypothesis was confirmed or rejected, based on significance levels.
This step allowed us to move from simple observation to more reliable, data-driven conclusions

Finally, a time-based analysis was performed by grouping data by year of release.

The average number of downloads per year was calculated to observe trends over time. This helped identify whether book popularity changes depending on how long the book has been available or the period in which it was released.

```
# 8. TIME ANALYSIS
# Average popularity by Year
yearly_trend = df.groupby('Year')['Downloads'].mean()
print("\n--- 7. Yearly Trends ---\n", yearly_trend.tail(5))
```

```
--- 7. Yearly Trends ---
 Year
2022    7416.715909
2023    5386.233010
2024    4973.061538
2025    5534.424658
2026    4040.884615
Name: Downloads, dtype: float64
```

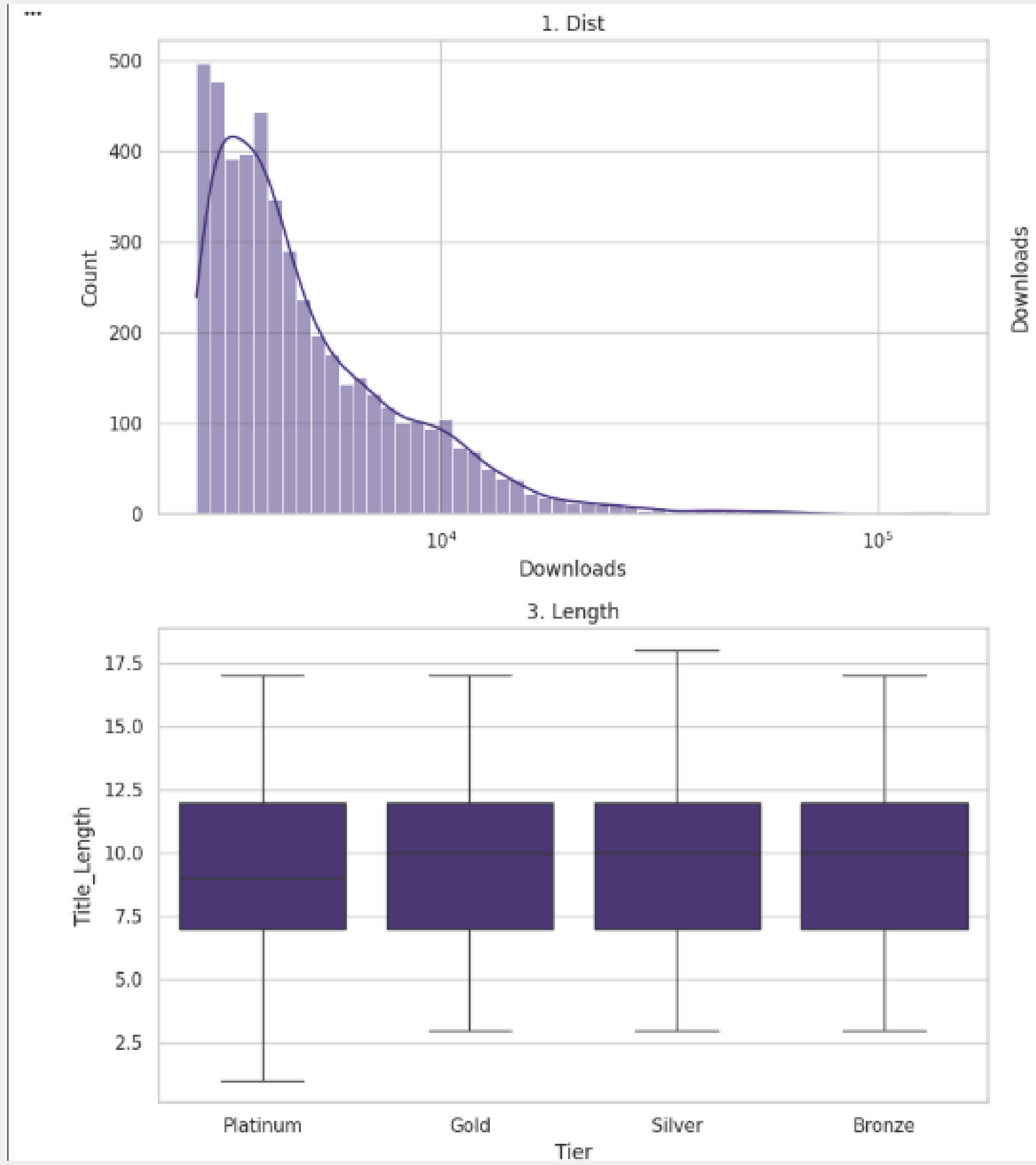
Overall, the analysis revealed key patterns in the dataset. Popularity is not evenly distributed, and a small number of books receive significantly higher downloads.

Relationships between variables exist, but they are not always strong. The hypothesis testing provided additional insights into how language, title length, and availability time influence book popularity.

This stage helped transform cleaned data into meaningful insights and supported further interpretation of the results.

Data Visualization

To better understand the dataset, multiple visualizations were created using Matplotlib, Seaborn, and Plotly. These visual tools help transform numerical data into clear and interpretable insights.
The analysis focuses on key aspects such as distribution, trends, relationships between variables, and popularity patterns.



The first set of charts focuses on general patterns in the data.

- “Is book popularity equally distributed among all books?”

This graph shows the distribution of downloads. The main idea is to understand whether most books receive similar attention or whether only a few books become extremely popular.
From the graph, we can see that book popularity is not evenly distributed. Most books have a relatively low or medium number of downloads, while only a small group of books receives a very high number of downloads. This means that the dataset has a strong imbalance: a few books dominate in popularity, while the majority remain less downloaded.
This is important because it shows that average values alone may not fully describe the dataset. Some extremely popular books can strongly influence the overall statistics.

- This boxplot compares title length (measured by the number of words) across different popularity tiers — Bronze, Silver, Gold, and Platinum.
- This visualization helps us identify whether there is a consistent difference between groups. For example, if higher tiers like Gold or Platinum show a higher median, it could suggest that more popular books tend to have longer titles.

2. Correlation Heatmap

Question:

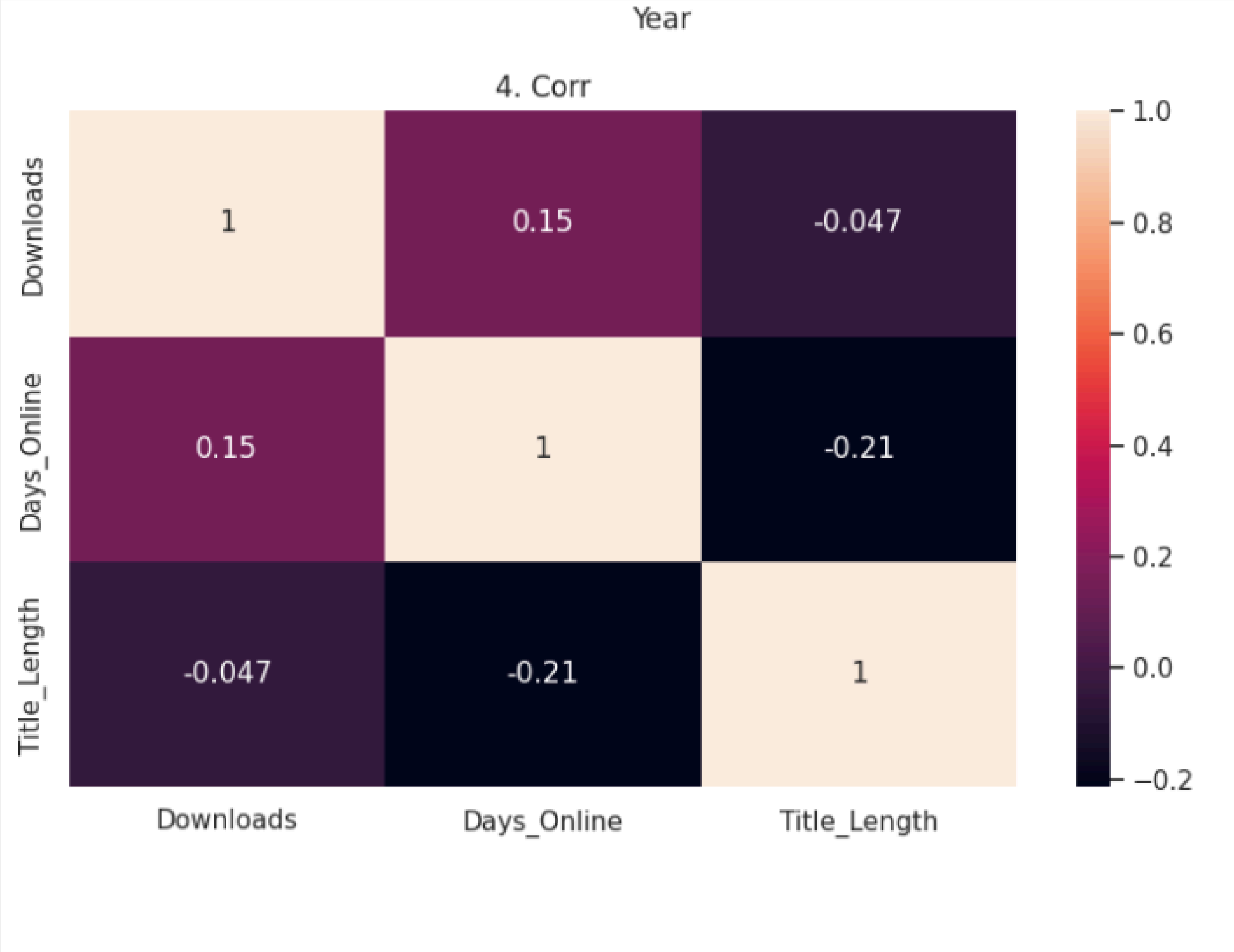
Which variables are connected with book popularity?

The correlation heatmap helps us understand the relationship between numerical variables: downloads, days online, and title length.

This graph is useful because it quickly shows whether one variable increases or decreases together with another variable. For example, if downloads and days online have a positive correlation, it means that books that have been available longer may have more downloads.

At the same time, the heatmap helps us check whether title length has any noticeable connection with downloads. If the correlation is weak, it means that the number of words in a title does not strongly affect popularity.

Overall, this graph gives a quick overview of which factors may be important and which ones are less influential.



3. Age vs Downloads Scatter Plot

Question:

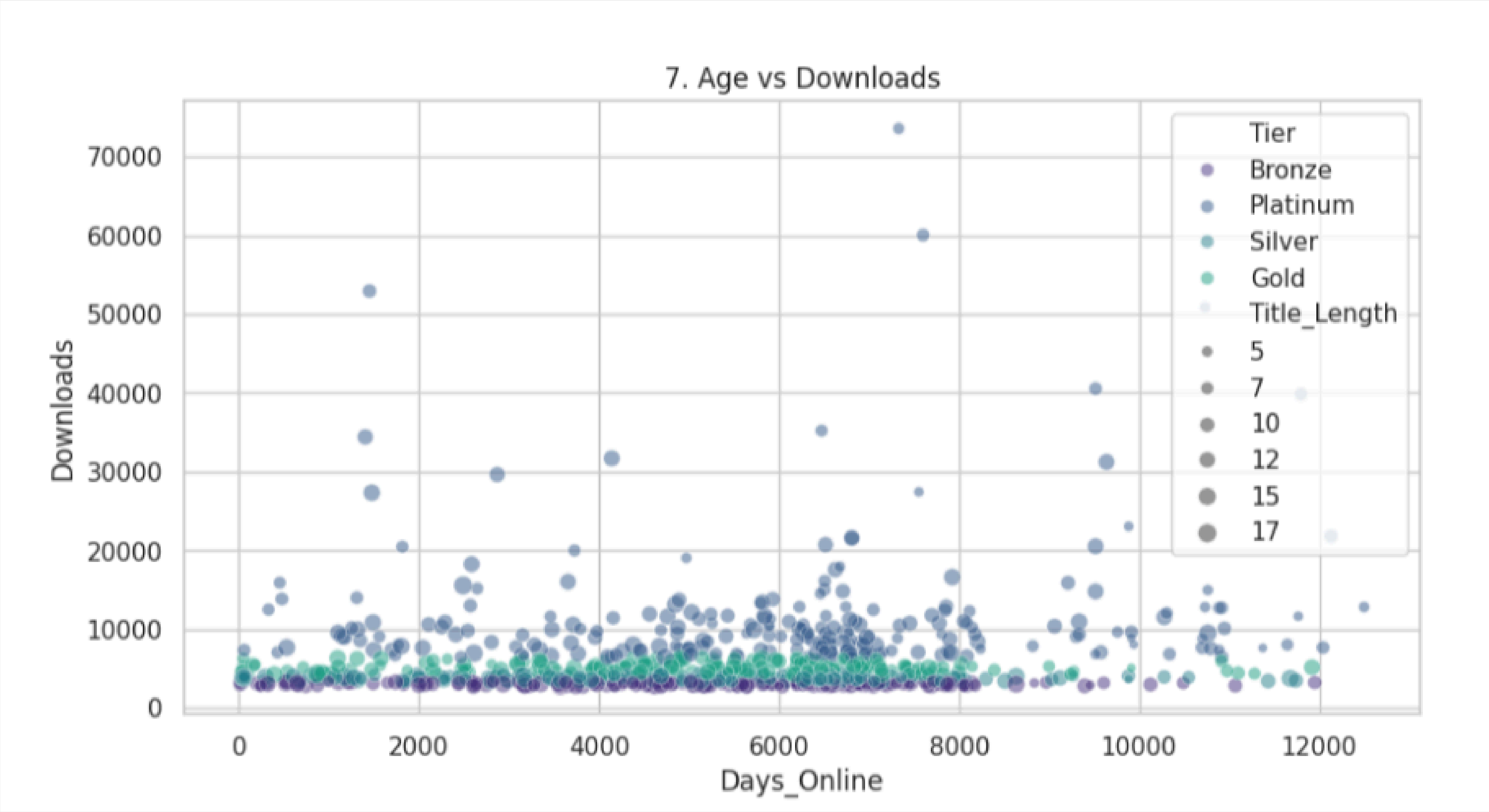
Do older books get more downloads over time?

This scatter plot compares days online with downloads. Each point represents one book from the dataset.

The graph helps us see whether books that have been available for a longer time tend to collect more downloads. If many points move upward as days online increases, it suggests that time may have a positive effect on popularity.

This visualization is especially interesting because it also includes popularity tiers and title length, so we can observe several variables at once. For example, we can see whether Platinum books are mostly older, or whether some newer books also become popular quickly.

This graph is useful because it shows patterns that are difficult to notice from tables alone.



4. Top 10 Authors

Question:

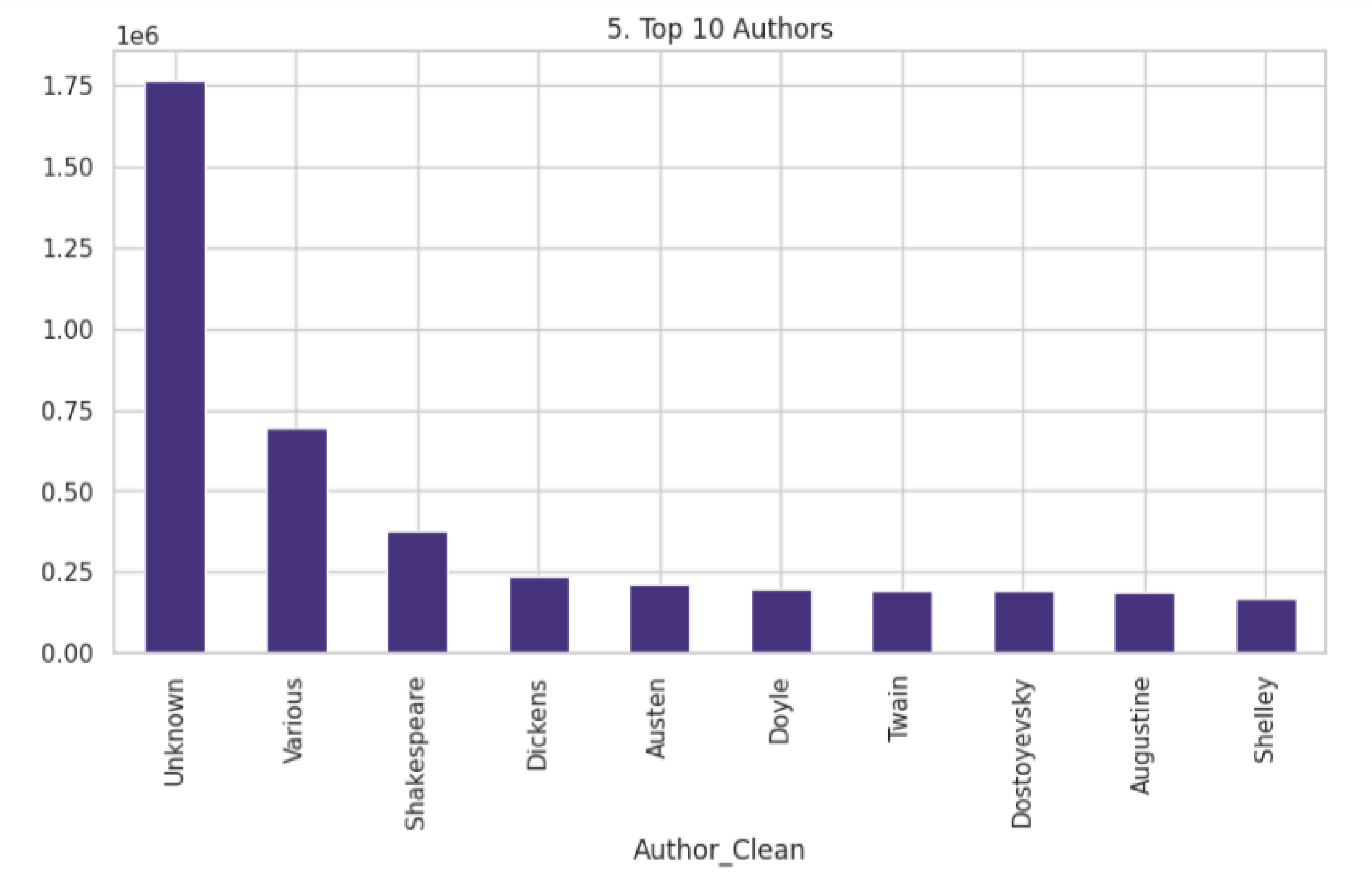
Which authors attract the most reader attention?

This bar chart shows the top 10 authors based on the total number of downloads.

The purpose of this graph is to identify which authors contribute the most to overall popularity on Project Gutenberg. Instead of looking only at individual books, this chart summarizes downloads by author.

From this visualization, we can see that some authors have a much stronger presence than others. This may be because they wrote famous classic books, have many works available, or are more commonly searched by readers.

This graph is important because it helps us understand popularity not only at the book level, but also at the author level. It shows which authors drive the biggest part of user interest in the dataset.



In this project, we collected, cleaned, and analyzed data from Project Gutenberg to better understand patterns in book popularity.
The results show that popularity is highly uneven — a small number of books receive the majority of downloads, while most books remain moderately popular. This confirms that the dataset follows a “long tail” distribution.
Our analysis also shows that **time plays an important** role. Books that have been available longer tend to accumulate more downloads. At the same time, factors like **title length do not have a strong impact**, and relationships between variables are generally weak or moderate.
Another key insight is the dominance of the **English language** and the influence of certain authors who generate a large share of total downloads.
Overall, this project demonstrates how raw data can be transformed into meaningful insights through data cleaning, statistical analysis, and visualization. It also shows that popularity is influenced by multiple factors, but no single variable fully explains it.